CLASS 8 · CBSE · GANITA PRAKASH PART 1 · DPP SHEET

Chapter 4: Quadrilaterals

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Three angles of a quadrilateral are 70°, 95° and 125°. Find the fourth angle.

2. In parallelogram PQRS, $\angle P = 72^\circ$. Find the other three angles.

3. In rectangle ABCD the diagonals meet at O and $\angle AOD = 50^\circ$. Find $\angle OAD$ and $\angle OAB$.

4. The diagonals of a rhombus are 10 cm and 24 cm. Find its side.

5. One angle of a rhombus is 70°. What angles does the shorter diagonal make with the sides?

6. In trapezium ABCD, $AB \parallel DC$, $\angle A = 68^\circ$ and $\angle B = 54^\circ$. Find $\angle D$ and $\angle C$.

7. In kite ABCD with $AB = BC$ and $CD = DA$, $\angle B = 120^\circ$ and $\angle D = 60^\circ$. Find $\angle A$.

8. Two equilateral triangles of side 4 cm are joined along a side. Name the quadrilateral and give its angles.

9. True or false: (i) every rhombus is a kite; (ii) a quadrilateral with perpendicular diagonals must be a rhombus.

10. The diagonals of a quadrilateral are equal, bisect each other and meet at 90°. What is it?

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Answer Key

1. 70°
2. $\angle Q = 108^\circ$, $\angle R = 72^\circ$, $\angle S = 108^\circ$
3. $\angle OAD = 65^\circ$, $\angle OAB = 25^\circ$
4. $\sqrt{5^2 + 12^2} = 13$ cm
5. 55° (the shorter diagonal bisects the 110° angles)
6. $\angle D = 112^\circ$, $\angle C = 126^\circ$
7. 90°
8. A rhombus with angles $60^\circ, 120^\circ, 60^\circ, 120^\circ$
9. (i) True. (ii) False; it could be a kite.
10. A square

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